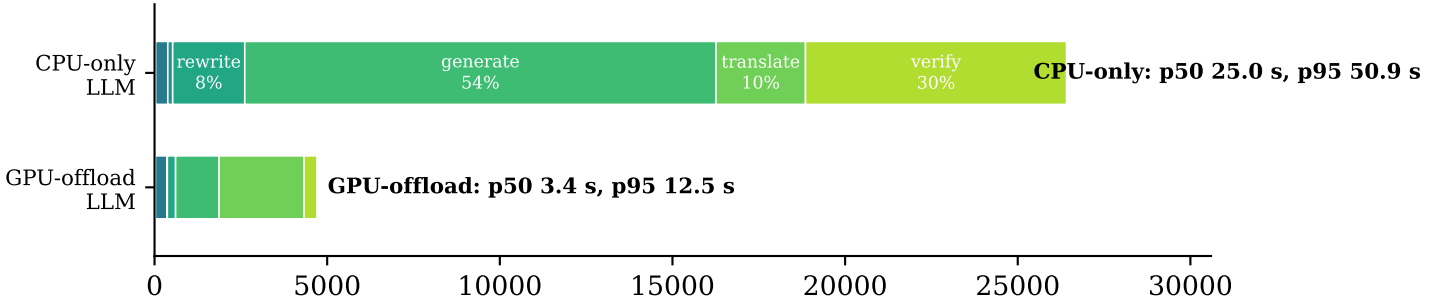
CPU-only
LLM
rewrite
8%
generate
54%
translate
10%
verify
30%
CPU-only: p50 25.0 s, p95 50.9 s
GPU-offload
LLM
GPU-offload: p50 3.4 s, p95 12.5 s
0
5000
10000
15000
20000
25000
30000
End-to-end per-stage latency (ms), mean over 37 queries × 3 runs. GPU offload cuts median latency 7.4× (25 s → 3.4 s).
normalize
kg_link
retrieve
rerank
grade
rewrite
generate
overhead